

NANRUI GONG

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EDUCATION

Johns Hopkins University

08/2021– 05/2025

B.S. in Applied Mathematics and Statistics, GPA: 3.92/4.0

Relevant Coursework: Data Structures, Operating Systems, Database, Machine Learning, Distributed Systems

SKILLS

Languages: Python, Swift, SQL, JavaScript, Julia, R, MATLAB, C++

Frameworks & Libraries: React, Node.js, Scikit-learn, Tensorflow, Numpy, SciPy, MLflow, FastAPI, Flask, REST API design, PyTest, JUnit

Tools: AWS (EC2, SQS, S3, Lambda, RDS), GCP (Cloud Run, Cloud SQL), FastAPI, Flask, MySQL, PostgreSQL, MongoDB, Docker, Kubernetes, Jenkins, Git, Databricks, Snowflake

PROFESSIONAL EXPERIENCE

Sciencia AI

Palo Alto, CA

Software Engineer Intern

July 2025–Present

- Engineered a **scalable** data ingestion pipeline that processes more than 1.2M user-generated reviews using Python and AWS services.
- Produced reviews into AWS **SQS**, created a Data Ingestion Service that ingests data from it, integrated **RDS**-stored user profile, and saved results into **S3** for downstream analytics.
- Created **Elasticache** to cache frequently accessed profile data, optimizing data integration and reducing end-to-end latency by 34%.
- Designed **Snowflake cloud data warehouse** schema to link product categories with reviews and users, optimizing for efficient querying by the downstream Machine Learning team.
- Created **NLP sentiment analysis** models with Scikit-learn and NLTK, designed and built a Query API Service with Flask to provide **RESTful API** endpoints for clients to retrieve analyzed results.

Dominion Energy

Richmond, VA

Software Engineer Intern

May 2024–August 2024

- Designed and developed a carbon emissions forecasting system using **Python** to replace legacy workflows and improved overall prediction performance.
- Integrated **Monte Carlo simulation** and **time-series models** to increase prediction accuracy and system performance, reduced forecast error by **5%** and improved runtime by **41.9%**.
- Built interactive dashboards in **React** and **Flask** to visualize forecasts for internal stakeholders.

PROJECTS EXPERIENCE

Full-Stack Task Management Web Application

- Designed and developed a **microservice-based** task management platform using **React** and **Node.js**, which is deployed on **AWS EC2**, applied auto-scaling for automatic horizontally scale.
- Implemented JWT-based authentication and role-based access control, configured **CI/CD pipeline** with Jenkins to ensure system robustness.
- Conducted comprehensive **testing** to validate reliability and performance across the system.

Cytometry Data Analysis Web App

- Designed and deployed a cloud-based web application on **GCP** for clinical biomarker analysis.
- Built a containerized **Flask API** and deployed it on Cloud Run, integrating with **Cloud SQL** for data storage.
- Automated statistical analysis **pipeline**, enabling researchers to perform Mann-Whitney U test through interface, reduced researcher testing time by around 20% compared with their old ways.

MLOps Pipeline for Digital Marketing Conversion Prediction

- Developed an end-to-end **MLOps pipeline** using FastAPI and deployed on **AWS Elastic Beanstalk**.
- Implemented a novel Discrete SPSA hyperparameter tuner for **Balanced Random Forest**, improving minority F1 score by 11.5%.
- Integrated automated data preprocessing, model training, evaluation, and **REST API** endpoints for **real-time predictions**.

Distributed Simulation Optimization Framework

- Developed a **distributed** simulation-optimization framework in Python and **Julia** for microgrid planning.

- Used **Docker** and **Kubernetes** to scale simulations across multiple nodes, achieving 31% faster convergence vs. baseline methods.